

Technical Requirements Specification

AI Environmental Scientist / Biodiversity Intelligence Agent

Implementation-ready requirements for a knowledge-grounded, multi-metric reasoning agent

Document Control	Value
Purpose	Define functional, technical, data, AI, API, evaluation, security, deployment, and acceptance requirements.
Primary source	Darukaa.Earth: AI Biodiversity Intelligence Chatbot Challenge (5-page challenge brief).
Target behavior	Behave like an AI environmental scientist rather than a generic chatbot.
Required modalities	Text input; structured input such as JSON. Geo/spatial context is a bonus.
Core architecture	Retrievable knowledge layer + conversational memory + multi-metric reasoning + evidence-backed recommendations.
Status	Technical requirements baseline

Source grounding note: Requirements marked as challenge-mandated are derived from the supplied challenge brief. Implementation details beyond the brief are engineering specifications proposed to make those requirements buildable and testable; they are not claims that the challenge explicitly mandates a particular framework, database, model, or cloud provider.

1. Executive System Definition

The system shall be an AI-powered conversational environmental intelligence agent that accepts a user's description of land/ecosystem conditions, retrieves relevant scientific and environmental evidence, reasons across multiple environmental variables, and produces specific, measurable, evidence-backed biodiversity recommendations. The supplied brief explicitly requires a retrievable knowledge layer rather than an LLM-only/prompt-only solution, multi-turn context, clarifying questions, and recommendations grounded in scientific evidence.

1.1 Non-negotiable capabilities

- Maintain a structured knowledge base covering soil health, land use/land cover, biodiversity indicators, climate factors, and human impact.
- Retrieve evidence from indexed research papers, reports, and/or environmental datasets using RAG, embeddings, vector search, structured retrieval, or a hybrid.
- Explain what evidence was retrieved and how it influenced the recommendation.
- Ask targeted clarifying questions when information required for a defensible recommendation is missing.
- Maintain conversation context across multiple turns and use prior user-provided environmental values in subsequent reasoning.
- Generate recommendations that connect at least three environmental variables together rather than treating variables independently.
- For every recommendation, provide the action, scientific rationale, impacted metrics, time horizon, and source/reference.
- Support free-form text and structured JSON-like input.
- Return structured, readable, actionable output.
- Avoid generic statements such as 'use sustainable practices'; recommendations must be operational and measurable.

1.2 Challenge-derived quality bar

The challenge evaluates reasoning depth at 30%, scientific grounding at 25%, knowledge-system design at 20%, conversational intelligence at 15%, and output clarity at 10%. It also explicitly rejects generic LLM-only solutions and shallow recommendations.

2. Scope and Use Cases

2.1 Primary use case

A user supplies environmental observations for a farm, parcel, ecosystem, or other land area. The agent normalizes the information, identifies missing variables, retrieves relevant scientific evidence, reasons over interactions, and returns prioritized interventions with measurable metrics and evidence.

2.2 Minimum supported scenario

Input	Example	Required processing
Soil organic carbon	0.3%	Validate range/unit; classify relative soil-carbon condition; use in soil–biodiversity reasoning.
Rainfall	Low	Normalize qualitative/quantitative value; connect to water availability and species survival.
Crop / land use	Monoculture wheat	Map to land-use/habitat context; assess diversification/fragmentation implications.
Region	Semi-arid	Use as spatial/ecological context; retrieve region-relevant evidence when available.

The challenge brief provides this exact type of example and expects recommendations such as agroforestry/intercropping, with explanation of effects on soil carbon and biodiversity and measurable improvement estimates, citing credible sources such as FAO or IPCC.

3. Target Technical Architecture

Recommended logical architecture:

- Client / API layer → accepts conversational text and structured environmental observations.
- Conversation Orchestrator → manages session state, intent, missing-data detection, tool routing, and response assembly.
- Environmental State Extractor → converts natural language and JSON into a canonical environmental state.
- Validation & Unit Normalizer → validates values, units, categorical vocabularies, and data quality.
- Knowledge Retrieval Layer → hybrid lexical/vector/metadata retrieval over scientific documents and environmental datasets.
- Evidence Reranker → ranks candidate evidence for relevance to the claim/recommendation and environmental context.
- Multi-Metric Reasoning Engine → constructs causal/associative links among soil, water/climate, land use, biodiversity, and human-impact variables.
- Recommendation Engine → proposes interventions and estimates expected metric direction/range where evidence supports it.
- Evidence Verifier / Citation Binder → ensures factual quantitative claims are traceable to retrieved evidence.
- Response Generator → converts validated recommendations into a structured answer with rationale, metrics, horizon, confidence, and references.
- Observability & Evaluation Layer → logs retrieval quality, missing-field questions, evidence coverage, latency, and test outcomes without exposing sensitive data.

3.1 High-level data flow

Step	Component	Output
1	User query	Text or structured JSON
2	Parse	Extract environmental entities, values, units, location, land-use context, and user goal
3	Validate	Range/unit/type validation; flag ambiguity
4	Assess completeness	Determine whether enough variables exist for multi-metric reasoning
5	Clarify if needed	Ask the smallest set of high-value questions
6	Retrieve	Retrieve scientific evidence and relevant dataset records
7	Reason	Connect ≥3 environmental variables and evaluate intervention trade-offs
8	Verify	Bind quantitative/scientific claims to evidence
9	Recommend	Generate concrete actions and expected metric impacts
10	Respond	Return structured recommendation + metrics + horizon + confidence + sources

4. Functional Requirements

ID	Requirement	Specification
FR-01	Text ingestion	Accept natural-language descriptions of environmental conditions and questions.
FR-02	Structured ingestion	Accept structured input (JSON or equivalent) containing environmental variables.
FR-03	Entity extraction	Extract soil, climate, land-use, biodiversity, pollution, deforestation, crop, region, and related entities.
FR-04	Unit normalization	Normalize units such as %, °C, mm, etc. and preserve original values for auditability.

FR-05	Missing-data detection	Identify variables needed to answer the user's question responsibly.
FR-06	Clarification	Ask concise, context-specific questions instead of guessing missing critical inputs.
FR-07	Session memory	Persist environmental facts and prior recommendations within the active conversation.
FR-08	Knowledge retrieval	Retrieve relevant scientific passages/records from the indexed knowledge layer.
FR-09	Evidence display	Expose source titles/organizations and relevant evidence context for recommendations.
FR-10	Multi-metric reasoning	Use at least three environmental variables in substantive recommendations.
FR-11	Recommendation generation	Produce concrete interventions with operational descriptions.
FR-12	Metric mapping	Map each recommendation to environmental metrics expected to change.
FR-13	Time horizon	Classify expected effects as short-, medium-, and/or long-term.
FR-14	Quantification	Provide improvement estimates only where supported by evidence; otherwise explicitly state that a reliable estimate is unavailable.
FR-15	Confidence	Optionally provide a confidence level based on evidence quality, retrieval relevance, and input completeness.
FR-16	Structured response	Return a predictable response schema suitable for UI rendering and automated testing.
FR-17	Traceability	Maintain a link from each evidence-backed claim to one or more retrieved sources.
FR-18	Failure handling	Handle insufficient evidence, contradictory sources, malformed inputs, and retrieval failures gracefully.

5. Canonical Environmental Data Model

Use a canonical internal representation so the reasoning engine does not operate directly on unstructured chat text.

Field	Type	Required	Validation / notes
session_id	string	Yes	Unique conversation identifier.
location	object/string	No	Region/place; coordinates are bonus capability.
soil.ph	number	No	Expected numeric pH; reject impossible values.
soil.organic_carbon_pct	number	No	Percentage; retain source and measurement date if supplied.
soil.moisture	number/object	No	Value plus unit or categorical state.
land_use	enum/string	No	e.g., monoculture, mixed cropping, forest, grassland; extensible taxonomy.
land_cover	enum/string	No	Normalized land-cover class if available.
biodiversity.species_richness	number	No	Count/index with unit/observation period if supplied.
biodiversity.habitat_diversity	number/object	No	Index/class plus methodology where available.
climate.temperature	number/object	No	Value, unit, time period.
climate.rainfall	number/object	No	Amount/class, unit, time

			period.
human_impact.pollution	object	No	Pollutant, level, unit, date where available.
human_impact.deforestation	object	No	Rate/extent/time period where available.
crop	string	No	Crop/species/system.
observed_problem	string	No	User-stated goal/problem.
data_provenance	array	Recommended	Source of each observed measurement.
measurement_date	date	Recommended	Important for climate and environmental observations.

5.1 Knowledge document schema

Field	Purpose
document_id	Stable identifier.
title	Paper/report/dataset title.
publisher_or_author	Source attribution.
publication_year	Temporal context.
source_url_or_identifier	Traceable source.
source_type	Research paper, report, dataset, etc.
domain_tags	Soil, biodiversity, rainfall, land use, etc.
region_tags	Geographic applicability.
evidence_text	Retrievable passage or record.
metric_claims	Structured quantitative/qualitative findings.
methodology_summary	Context for interpreting findings.
limitations	Applicability/uncertainty notes.
embedding	Vector representation for semantic retrieval.
lexical_index	BM25/TF-IDF or equivalent lexical retrieval representation.

6. Knowledge Retrieval and RAG Requirements

The knowledge layer is critical: the challenge explicitly expects RAG, embeddings, vector databases, or structured datasets, indexing research papers/reports/environmental datasets, and clearly showing how knowledge is retrieved and used.

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6.1 Recommended retrieval pipeline

1. Build a domain-aware query from the user's environmental state rather than using the raw question alone.
2. Retrieve candidates through hybrid lexical + semantic search.
3. Apply metadata filters for domain, region, land-use type, metric, publication year, and evidence type when applicable.
4. Rerank candidates using query relevance plus environmental-context overlap.
5. Return top evidence passages with document IDs and source metadata.
6. Pass only the most relevant evidence into the reasoning/generation stage to reduce hallucination opportunities.
7. Record retrieved IDs and scores for reproducibility and evaluation.

6.2 Retrieval quality requirements

Metric	Requirement
Evidence relevance	Top retrieved passages must directly address the environmental relationship or intervention being reasoned about.
Source traceability	Every displayed scientific claim must map to a source record.
Context matching	Region/climate/land-use differences must be considered before transferring quantitative estimates.
Contradiction handling	If credible sources disagree, surface the uncertainty instead of silently selecting one.
No-evidence behavior	If evidence is insufficient, state that a reliable conclusion/estimate cannot be supported.

7. Multi-Metric Reasoning Engine

This is the core differentiator in the challenge: the system must connect soil health ↔ biodiversity, water availability ↔ species survival, and land use ↔ habitat fragmentation, rather than producing single-variable answers. The brief explicitly requires at least three environmental variables together.

7.1 Reasoning representation

Represent each case as a small environmental graph:

Node type	Examples
Observed state	Low rainfall, SOC 0.3%, wheat monoculture
Environmental mechanism	Low soil carbon → reduced soil function; monoculture → lower habitat diversity
Constraint	Water limitation, regional climate, available land
Intervention	Cover crops, intercropping, agroforestry, habitat strips, etc.
Target metric	SOC, soil moisture, species richness, habitat diversity
Evidence	Study/report/dataset supporting relationship or intervention

7.2 Minimum reasoning contract

- Identify at least three relevant variables from the case.
- State the relationships/mechanisms linking those variables.
- Generate candidate interventions compatible with the constraints.
- Check each candidate against retrieved evidence.
- Map each candidate to measurable metrics and a realistic time horizon.
- Avoid quantitative estimates when the evidence is not transferable to the user's context.
- Explain uncertainty and assumptions.

8. Recommendation Engine

Every recommendation shall contain the elements explicitly required by the challenge: what to do, why it works, which environmental metric improves, and a reference to a study/report/model.

Output field	Required content
action	Specific intervention and implementation description.
rationale	Mechanistic/scientific explanation tied to the user's conditions.
metrics_impacted	Primary and secondary environmental metrics.
direction_of_change	Increase/decrease/maintain where evidence supports it.
estimated_effect	Numeric range only if supported; otherwise qualitative.
time_horizon	Short / medium / long term.
evidence	Source(s) supporting the intervention and/or estimate.
applicability	Why the evidence is relevant to this case.
assumptions	Important assumptions or missing information.
confidence	Optional but recommended; based on evidence and input quality.
tradeoffs	Potential negative effects, resource costs, or competing outcomes.
measurement_plan	How the user can observe whether the intervention is working.

8.1 Recommendation quality rules

- Do not output vague recommendations without an action mechanism.
- Prefer interventions that address several observed constraints simultaneously.
- Do not fabricate percentages, timelines, citations, study names, or causal relationships.
- Distinguish evidence-supported effect sizes from system-generated hypotheses.
- When evidence is only observational/correlational, do not describe it as proven causation.
- Include a measurement plan so recommendations can be evaluated in the user's setting.

9. Conversational Intelligence and Memory

The agent must ask clarifying questions when inputs are incomplete, support multi-turn conversations with memory, and adapt responses to prior context. The brief illustrates this with a biodiversity-decline query followed by questions for soil organic carbon, rainfall pattern, and land-use type.

9.1 Clarification policy

- Ask only for information that materially changes the recommendation or its evidence applicability.
- Prioritize variables needed to satisfy the three-variable reasoning requirement.
- Do not repeatedly ask for values already provided in the conversation.
- If a missing value can safely be represented as unknown, do not force the user to provide it.
- If the user declines to provide a value, proceed with explicit assumptions and reduced confidence where possible.

9.2 Session memory schema

Memory item	Example
User goal	Increase biodiversity on agricultural land
Environmental facts	SOC 0.3%; rainfall low; wheat monoculture
Location context	Semi-arid region
Prior recommendations	Intercropping suggested
User feedback	Cannot irrigate additional area
Evidence already retrieved	Document IDs and key passages
Open questions	No biodiversity baseline provided

10. API and Interface Requirements

10.1 POST /api/v1/agent/query

Purpose: submit a conversational or structured environmental query.

Request example:

```
{
  "session_id": "abc123",
  "message": "Biodiversity is declining on my land.",
  "environment": {
    "soil": {"organic_carbon_pct": 0.3},
    "climate": {"rainfall": "low"},
    "crop": "wheat",
    "land_use": "monoculture",
    "region": "semi-arid"
  }
}
```

10.2 Response contract

```
{
  "status": "clarification_required | recommendation_ready | insufficient_evidence",
  "summary": "...",
  "missing_information": [],
  "reasoning": {
    "variables_used": [],
    "relationships": []
  },
  "recommendations": [
    {
      "action": "...",
      "why_it_works": "...",
      "metrics_impacted": [],
      "time_horizon": "short|medium|long",
      "estimated_effect": "...",
    }
  ]
}
```

```

    "confidence": "...",
    "evidence": [
      {"document_id": "...", "title": "...", "source": "..."}
    ],
    "assumptions": [],
    "measurement_plan": "..."
  }
}

```

10.3 Additional endpoints

Endpoint	Purpose
POST /api/v1/session	Create a conversation session.
GET /api/v1/session/{id}	Retrieve active session environmental state and conversation metadata.
POST /api/v1/knowledge/search	Developer/evaluation endpoint for testing retrieval independently.
GET /api/v1/health	Liveness/readiness check.
GET /api/v1/evidence/{document_id}	Retrieve source metadata and indexed evidence passage for traceability.

11. Suggested Implementation Stack

The challenge does not prescribe a particular technology stack. The following is an implementation baseline chosen for simplicity, explainability, and hackathon delivery speed.

Layer	Suggested technology	Reason
Backend	Python + FastAPI	Simple typed API and rapid integration with NLP/ML libraries.
LLM	Provider-agnostic instruction-following model	Generation/orchestration; should not be the sole knowledge source.
Embeddings	Provider or open-source embedding model	Semantic retrieval.
Vector store	PostgreSQL + pgvector or dedicated vector DB	Persistent vector retrieval with metadata.
Lexical retrieval	BM25/TF-IDF	Exact terminology and scientific keyword matching.
Document processing	PDF/text extraction + chunking pipeline	Build searchable evidence corpus.
Metadata store	PostgreSQL	Structured environmental data, sources, sessions, evaluation records.
Validation	Pydantic	Typed request/response and environmental-state validation.
Evaluation	Python test suite + retrieval/reasoning benchmark	Automated quality checks.
Deployment	Docker	Reproducible local/demo environment.
Frontend	Minimal web chat	Only enough UI to demonstrate agent behavior; UI is not the primary challenge focus.

12. AI Agent Orchestration Requirements

12.1 Agent state machine

State	Entry condition	Exit condition
UNDERSTAND	New user message	Intent + environmental entities extracted
VALIDATE	Environmental state available	Values normalized and validity checked
CLARIFY	Critical information missing	User supplies information or accepts assumptions
RETRIEVE	Sufficient context	Evidence candidates returned
REASON	Evidence available	Multi-metric relationships and candidate actions produced
VERIFY	Candidate recommendation exists	Claims mapped to evidence; unsupported

		claims removed/qualified
RESPOND	Verified recommendation	Structured response returned
LEARN_CONTEXT	Response completed	Updated session state saved

12.2 Prompt/tool separation

- The LLM should orchestrate and explain; deterministic code should handle schema validation, unit normalization, retrieval metadata, and response validation.
- Scientific evidence should be supplied to the model as retrieved context rather than relying on model memory.
- The model should be instructed to abstain from unsupported quantitative claims.
- The final response validator should reject recommendations lacking action, rationale, metric mapping, time horizon, or evidence where evidence is required.

13. Evaluation and Acceptance Tests

The test suite should directly mirror the challenge's evaluation dimensions.

Area	Test	Acceptance criterion
Reasoning depth	Case contains low rainfall + low SOC + monoculture.	Recommendation explicitly connects ≥ 3 variables and explains the mechanism.
Scientific grounding	Recommendation contains a quantitative claim.	Claim has a traceable source and applicability context.
Knowledge system	Disable retrieval.	System detects/handles missing evidence rather than behaving as if evidence was retrieved.
Conversational	Omit key variable on turn 1, provide it on turn 2.	Agent retains earlier values and integrates the new value without repetition.
Clarification	Give vague biodiversity-decline statement.	Agent asks targeted questions for materially important missing variables.
Output clarity	Run complete case.	Response contains recommendation, impacted metrics, time horizon, and references.
Robustness	Malformed JSON / invalid units.	Validation error is clear and does not crash the agent.
Contradiction	Retrieve conflicting evidence.	Agent reports uncertainty/conflict rather than inventing consensus.
No hallucination	Ask for an unsupported effect size.	Agent declines to provide a fabricated number.
Traceability	Inspect recommendation.	Evidence IDs/source metadata can be followed back to indexed records.

13.1 Suggested quantitative engineering KPIs

KPI	Target for MVP
Retrieval Recall@5	≥ 0.80 on a manually labeled benchmark set.
Evidence citation coverage	$\geq 95\%$ of externally verifiable scientific claims have a source.
Unsupported numeric-claim rate	$< 2\%$ on evaluation set.
Multi-metric compliance	$\geq 95\%$ of applicable recommendation cases use at least 3 environmental variables.
Clarification precision	$\geq 90\%$ of asked questions judged materially useful.
Structured response validity	$\geq 99\%$ schema-valid responses in automated tests.
Demo response latency	Target < 10 seconds for normal retrieval + generation path, subject to model/provider latency.

These engineering targets are proposed implementation goals, not challenge-stated scoring thresholds.

14. Data Ingestion and Corpus Construction

15. Collect authoritative research papers, reports, and environmental datasets relevant to the five required knowledge domains.
16. Store source metadata before chunking so every chunk remains traceable.

17. Chunk documents by semantic section/paragraph rather than arbitrary fixed lengths alone.
18. Attach domain, metric, region, date, source type, and methodology metadata to chunks where available.
19. Generate embeddings for semantic search and a lexical index for exact terminology.
20. Run a duplicate/near-duplicate check to reduce repeated evidence.
21. Record ingestion version so evaluation results can be reproduced.
22. Do not treat search-engine snippets or unverified web text as scientific evidence without source validation.

14.1 Required knowledge domains

Domain	Minimum coverage
Soil health	pH, organic carbon, moisture
Land use / land cover	Land-use classes, habitat context, fragmentation-related evidence
Biodiversity	Species richness, habitat diversity
Climate	Temperature, rainfall
Human impact	Pollution, deforestation

These domains and metrics are explicitly listed in the challenge brief. [filecite?turn0file0?L15-L22?](#)

15. Security, Reliability, and Governance

- Treat user-supplied text and structured values as untrusted input.
- Validate and constrain tool/API parameters before execution.
- Do not allow retrieved documents to override system instructions; treat retrieved content strictly as evidence.
- Keep secrets/API keys in environment variables or a secret manager, never in source control.
- Sanitize logs so personal or sensitive user data is not unnecessarily retained.
- Version prompts, retrieval configuration, embedding model, corpus, and recommendation rules for reproducibility.
- Expose uncertainty and evidence limitations instead of presenting unsupported environmental predictions as facts.
- Provide clear distinction between measured user inputs, retrieved evidence, inferred relationships, and recommendations.

16. Observability and Debugging

Event	Fields to log
Query received	session_id, timestamp, input mode
Extraction	entities/fields detected, validation errors
Clarification	missing fields, question generated
Retrieval	query representation, filters, document IDs, scores, latency
Reasoning	variables used, candidate interventions, evidence IDs
Verification	claims checked, unsupported claims removed/qualified
Response	schema status, recommendation IDs, total latency
Feedback	user feedback category if collected

Logs should support debugging and evaluation without becoming a hidden substitute for evidence traceability.

17. Deployment and Repository Requirements

The submission must be runnable and reviewable. The challenge requires one Word document containing the GitHub repository link, live demo URL where applicable, a brief README covering architecture/database-schema/local setup/CI-CD, and any additional credentials or notes needed to run the project. [filecite?turn0file0?L107-L126?](#)

Artifact	Requirement
README.md	Architecture diagram/description, database/schema, local setup, environment variables, run commands, CI/CD details.
Source code	Organized backend/agent/retrieval/data/evaluation modules.
Knowledge corpus manifest	List of indexed sources and versions.
Environment example	.env.example without secrets.

Docker	Dockerfile and/or compose configuration for reproducible setup.
Tests	Unit, integration, retrieval, and end-to-end cases.
Demo	Live URL if available; otherwise clear local execution instructions.
Submission docx	GitHub link, demo URL, README overview, credentials/notes as required.

18. Recommended Repository Structure

darukaa-biodiversity-agent/

```
├── app/
│   ├── api/
│   ├── agent/
│   ├── extraction/
│   ├── retrieval/
│   ├── reasoning/
│   ├── recommendations/
│   ├── evidence/
│   ├── memory/
│   └── schemas/
├── data/
│   ├── raw/
│   ├── processed/
│   └── manifests/
├── scripts/
│   ├── ingest.py
│   └── evaluate.py
├── tests/
│   ├── unit/
│   ├── retrieval/
│   └── e2e/
├── docs/
│   └── architecture.md
├── Dockerfile
├── docker-compose.yml
├── requirements.txt / pyproject.toml
├── .env.example
└── README.md
```

19. MVP vs. Stretch Scope

MVP — required to satisfy challenge	Stretch — improves sophistication
Text + JSON input	Geo-coordinates and spatial retrieval
Five knowledge domains	More environmental domains
Hybrid/vector/structured retrieval	Knowledge graph / graph database
Multi-turn memory	Long-term user/project memory
≥3-variable reasoning	Explicit causal graph or probabilistic model
Evidence-backed recommendations	Scenario simulation and intervention comparison
Source traceability	Interactive evidence explorer
Short/medium/long horizon	Time-series forecast of metrics
Confidence/uncertainty	Formal uncertainty calibration
Minimal UI	Map-based environmental dashboard

20. Final Acceptance Checklist

- ☐ Knowledge layer exists and is retrievable; the solution is not LLM-only.
- ☐ Indexed sources include research papers/reports and/or environmental datasets.
- ☐ Soil health covers pH, organic carbon, and moisture.

- ☐ Land use/land cover is represented.
- ☐ Biodiversity covers species richness and habitat diversity.
- ☐ Climate covers temperature and rainfall.
- ☐ Human impact covers pollution and deforestation.
- ☐ Text input works.
- ☐ Structured JSON input works.
- ☐ Agent asks clarifying questions when required inputs are missing.
- ☐ Agent remembers environmental context across turns.
- ☐ Recommendations use at least three environmental variables.
- ☐ Each recommendation says what to do.
- ☐ Each recommendation explains why it works.
- ☐ Each recommendation identifies impacted metrics.
- ☐ Each recommendation provides scientific evidence/reference.
- ☐ Output includes a time horizon.
- ☐ Quantitative estimates are evidence-backed and contextualized.
- ☐ Retrieval and evidence usage can be demonstrated to reviewers.
- ☐ The project includes automated tests and a reproducible local setup.
- ☐ README includes architecture, database/schema, setup, and CI/CD details.
- ☐ Submission document contains repository link and live demo URL where applicable.

21. Direct Alignment With the Challenge Brief

Challenge requirement	Technical implementation in this specification
Retrievable knowledge layer	Hybrid/vector/structured retrieval architecture and indexed corpus.
Environmental metrics	Canonical data model for soil, land use/cover, biodiversity, climate, and human impact.
Clarifying questions + memory	Conversation state machine, completeness checks, and session memory.
Evidence-backed recommendations	Recommendation schema + evidence verifier + citation binding.
Multi-metric reasoning	Environmental graph and minimum ≥ 3 -variable reasoning contract.
Text + structured input	API supports free-form message and environment JSON.
Clear outputs	Recommendation, metrics, horizon, confidence, evidence, assumptions, measurement plan.
Evaluation	Test plan mapped to reasoning, grounding, knowledge, conversation, and clarity.
Submission	README/repository/demo/submission-document requirements.

End of Technical Requirements Specification